

# KYLE WILLIAMSON

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## SUMMARY

Software engineering student building production systems from mobile apps to microservices, with active interest in low-level and systems programming. Seeking a backend, full-stack, or systems-oriented new-graduate role.

## EDUCATION

### York University – Lassonde School of Engineering

Sept 2021 – April 2027

*Bachelor of Engineering, Software Engineering* **Relevant Coursework:** Mission-Critical Systems, Software Design & Testing, Data Structures & Algorithms, Operating Systems, Computer Organization, Embedded Systems, Capstone Engineering Project

## PROJECTS

### BlockBid Auction System – Full-Stack Microservices Platform

Oct 2025 – Dec 2025

*Docker / Spring Boot / WebSocket / Ethereum*

- Architected 6-service microservices auction platform with database-per-service pattern, real-time WebSocket bidding with JWT/RBAC authentication, Ethereum blockchain audit trails, and 30+ REST endpoints, deployed via Docker Compose with a single command.

### Runz – Street Basketball Ranking Mobile App

Jan 2026 – Present

*Flutter / Supabase / Riverpod / PostGIS / Deno*

- Architected full-stack competitive 1v1 basketball platform targeting 6 platforms (iOS, Android, Web, macOS, Linux, Windows) with real-time game sessions via Supabase Realtime WebSocket subscriptions, eliminating polling for live status updates across 12+ Postgres tables during active matches.
- Designed dual-implementation Elo ranking engine (Dart + PL/pgSQL) with dynamic K-factors, 5-tier progressive ranking system, GPS verification weighting, and anti-cheat protections including streak cooldowns, rep quarantine, and server-side proximity validation via PostGIS.
- Built 3 Edge Functions (TypeScript/Deno) supporting 15+ push notification types via OAuth2-signed FCM v1 API, persistent offline action queue with exponential-backoff retry, and cron-driven auto-resolution for abandoned game sessions, backed by unit and widget tests.

### YorkU Conference Room Scheduler

Sept 2025 – Dec 2025

*Java / Design Patterns / UML*

- Led team of 4 in building enterprise room booking system with conflict-free scheduling, role-based access control, automated waitlisting, and comprehensive UML documentation, applying Observer, Factory, and Strategy design patterns.

### GTA Real Estate Hotspots – Spatial Network Analysis

Sept 2025 – Dec 2025

*Python / PyTorch / NetworkX*

- Analyzed 358,713 building permits from Toronto Open Data using graph-based spatial network of 98 regions and 165 connections, with Spatial Autoregressive model achieving 44.7% lower RMSE ( $p = 0.037$ ) and 60% precision in identifying top-10 real estate growth hotspots.

### Terminal Portfolio – Interactive Developer Website

April 2025

*Vanilla JS / Web Audio API / CSS3*

- Built interactive terminal emulator with custom shell (14+ commands, arrow-key history, inactivity-triggered auto-demo, recruiter skip-path via URL parameter), procedural Web Audio API sound engine synthesizing POST beeps and keystroke clicks from oscillator nodes with zero audio files, and CRT-era aesthetic via layered CSS gradients and keyframe animations, all vanilla JS, HTML5, CSS3, zero dependencies.

## PROFESSIONAL EXPERIENCE

### Aritzia – Risk Associate

Nov 2025 – Feb 2026

*Vaughan, ON*

- Conducted inventory audits and fraud monitoring across store operations, coaching team members on loss prevention procedures and inventory accuracy.

### Northern Shield Security – Security Professional

Nov 2024 – Present

*Richmond Hill, ON*

- Performed threat assessments and maintained incident documentation for high-value retail environments, coordinating security response with mall security teams.

### Apple – Sales Specialist

Oct 2023 – April 2024

*Toronto, ON*

- Translated complex technical concepts for 30–50 customers daily across Apple’s product ecosystem while completing ongoing technical certification training.

## SKILLS

**Languages:** Python, Java, C/C++, JavaScript, TypeScript, Dart, SQL, SystemVerilog, HTML/CSS

**Frameworks:** Spring Boot, React.js, Flutter, Riverpod, Streamlit, Node.js, Express.js

**Data & ML:** PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, NetworkX, Jupyter Notebooks

**Infrastructure:** Git, Docker, Docker Compose, Maven, Linux/macOS, CI/CD (GitHub Actions)

**Databases:** SQLite, MySQL, PostgreSQL, Supabase

**APIs & Protocols:** REST APIs, WebSocket, SSE, JWT, OAuth, Discord/Telegram Bot APIs

**Tools:** Postman, Agile, JUnit, Playwright, Selenium, Excel/Numbers/Sheets

**Spoken Languages:** French, Italian